ZERO SPEED FOR EXCITED RANDOM WALKS ON THE EVENT OF NON-TRANSIENCE

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ABSTRACT. An excited random walk on $\mathbb{Z}^d$ moves, on its i-th visit to a site x, according to the i-th entry of a stack of transition probabilities attached to x; the stacks are independent and identically distributed over the sites. Kosygina and Zerner asked whether $X_n/n \to 0$ almost surely on the event that $|X_n|$ does not tend to infinity, assuming only that the stacks are i.i.d. and every entry is positive. We prove that it does, in every dimension and with no further restriction on the stacks, which may be infinite and may push in any direction. We also prove the excited-walk form of Zerner's law of large numbers for random walk in random environment: if, for a coordinate direction e, the walk tends to $+\infty$ and to $-\infty$ in direction e with probability zero, then $X_n \cdot e/n \to 0$ almost surely. The strong Markov restart in Zerner's argument is replaced by a sum over the finitely many paths the walk can follow between its first and last visits to a level.

1. INTRODUCTION

Let $E = \{\pm e_1, \dots, \pm e_d\}$ be the unit vectors of $\mathbb{Z}^d$. A *cookie environment* is a family $\omega = (\omega(x,a,i))_{x\in\mathbb{Z}^d,\, a\in E,\, i\geq 1}$ with $\omega(x,a,i) \geq 0$ and $\sum_{a\in E}\omega(x,a,i) = 1$. The sequence $\omega(x,\cdot,\cdot)$ is the *stack* at x. Given ω and $y \in \mathbb{Z}^d$, the excited random walk $(X_n)_{n\geq 0}$ starts at $X_0 = y$ and moves by

$$P_{y,\omega}\big(X_{n+1} = X_n + a \,\big|\, X_0, \dots, X_n\big) = \omega\big(X_n, a, \#\{0 \leq m \leq n : X_m = X_n\}\big), \qquad a \in E, \tag{1}$$

so on its i-th visit to a site the walk uses the i-th entry of that site's stack. The environment is random with law $\mathbb{P}$, and $P_y = \mathbb{E}[P_{y,\omega}]$ is the annealed law. We use the two assumptions of the survey of Kosygina and Zerner [6]:

(IID) the stacks $\omega(x,\cdot,\cdot)$, $x \in \mathbb{Z}^d$, are independent and identically distributed under $\mathbb{P}$;
(EL) $\mathbb{P}$-almost surely, $\omega(x,a,i) > 0$ for all x, a and i.

Nothing else is assumed. There is no uniform lower bound in (EL), the entries of one stack need not be independent, and a stack may have infinitely many entries that differ from the simple random walk step, with drifts of either sign.

For $\ell \in \mathbb{R}^d \setminus \{0\}$ let $A_\ell = \{X_n \cdot \ell \to +\infty\}$, and let

$$N = \big\{|X_n| \not\to \infty\big\}$$

be the event that the walk does not go to infinity, which we call non-transience. Kosygina and Zerner asked the following [6, Problem 4.7].

Problem. Assume (IID) and (EL), and let $\ell \in \mathbb{R}^d \setminus \{0\}$. Does $X_n \cdot \ell/n$ converge to 0 P_0-almost surely on N?

Date: November 2026.

2020 *Mathematics Subject Classification.* 60K35, 60K37, 60F15.

Key words and phrases. excited random walk, cookie random walk, law of large numbers, self-interacting random walk, zero-one law.

For random walk in an i.i.d. elliptic random environment, which is the case in which $\omega(x, a, i)$ does not depend on i, Zerner [9, Theorem 1] proved the following: if $e \in E$ and $P_0(A_e \cup A_{-e}) = 0$, then $X_n \cdot e/n \to 0$ P_0-almost surely. In dimension one, Amir, Berger and Orenshtein [1, Theorem 1.3] proved that X_n/n converges almost surely to a deterministic limit for stationary ergodic elliptic stacks on $\mathbb{Z}$, infinite stacks included; when $P_0(N) > 0$ the limit is zero, since on N the walk visits some site infinitely often. We answer the problem in every dimension.

Theorem 1.1. *Let $d \geq 1$ and assume (IID) and (EL). Then $X_n/n \to 0$ P_0-almost surely on N. If $P_0(N) > 0$, then $X_n/n \to 0$ P_0-almost surely.*

The main step is the following one-sided statement, which holds for every coordinate direction separately.

Proposition 1.2. *Assume (IID) and (EL), and let $e \in E$. If $P_0(A_e) = 0$, then $\limsup_n X_n \cdot e/n \leq 0$ P_0-almost surely.*

Applied to e and to $-e$, Proposition 1.2 gives the excited-walk form of Zerner's theorem.

Corollary 1.3. *Assume (IID) and (EL), and let $e \in E$. If $P_0(A_e \cup A_{-e}) = 0$, then $X_n \cdot e/n \to 0$ P_0-almost surely.*

When the stacks are constant in i this is [9, Theorem 1]. When $P_0(N) > 0$, the zero-one law [6, Theorem 3.16] gives $P_0(A_e \cup A_{-e}) = 0$ for every $e \in E$, and Theorem 1.1 follows; the deduction is in Section 5.

The idea of the proof. Zerner's proof has two parts. Fix $e \in E$, call $\{x : x \cdot e = m\}$ level m, let T_m be the first time the walk is at level m, and, for an integer $L \geq 1$, let $h_{m,L}$ be the time from T_m to the last visit to level m before T_{m+L}. The first part concerns a single path [9, Lemma 3]: if $\limsup_n X_n \cdot e/n > 0$, then for some integer c and every L the levels m with $h_{m,L} \leq c$ have positive upper density. The second part bounds the probability that $h_{m,L} \leq c$. At the last visit to level m the walk is at one of finitely many sites z, for the r-th time with $r \leq c$, and by the strong Markov property its law from then on is $P_{z,\omega}$; the probability that it then reaches level $m + L$ without returning to level m involves only the environment at levels $\geq m$, which the walk has not seen at time T_m.

For an excited walk the second part fails. After the r-th visit to z the law of the walk depends on which entries of the stacks at z, and at the other sites visited since T_m, have been used, and this is not a function of z and r. We keep the whole path between T_m and the last visit instead. Let Π_c be the set of nearest-neighbour paths $p = (p_0, \ldots, p_s)$ with $p_0 = 0$, $s \leq c$ and $p_s \cdot e = 0$, which is finite, and let ω^p be the environment obtained from ω by deleting from each stack the entries used along p. Bounding the probability of following p by one, we obtain under (IID) alone, for all $m \geq 0$ and $L > c$,

$$P_0\big(T_{m+L} < \infty,\ h_{m,L} \leq c \,\big|\, X_0, \ldots, X_{T_m}\big) \leq \sum_{p \in \Pi_c} \mathbb{E}\Big[P_{p_s,\omega^p}\big(T_L < \infty,\ X_n \cdot e > 0 \text{ for } 1 \leq n \leq T_L\big)\Big]$$

on $\{T_m < \infty\}$, where T_L is the first time the walk started at p_s is at level L (Lemma 3.2). The right side depends neither on m nor on the path up to time T_m. If $P_0(A_e) = 0$, it tends to 0 as $L \to \infty$ for each c (Lemma 3.1), so the levels with $h_{m,L} \leq c$ have upper density at most this bound, and Zerner's lemma gives Proposition 1.2.

Related work. Excited random walk was introduced by Benjamini and Wilson [3], with one cookie of drift in direction e_1 at each site. Zerner [10] and Menshikov, Popov, Ramírez and Vachkovskaia [7] studied walks whose drifts point into a half-space, and Bérard and Ramírez [2] proved a central limit theorem for the Benjamini–Wilson walk; in these models the walk is

transient and $P_0(N) = 0$. The laws of large numbers of Holmes [4] and Pham [8] in high dimensions assume simple transverse steps. Theorem 4.8 of the survey [6] proves a law of large numbers under the assumption $P_0(A_{\ell_i} \cup A_{-\ell_i}) = 1$ for a basis $\ell_1, \dots, \ell_d$, which the survey relates to Problem 4.7; Corollary 1.3 covers a coordinate direction in which this probability is 0. In dimension one, besides [1], infinite stacks were treated by Kosygina and Peterson [5] for Markovian stacks. A table of notation is in Appendix A.

2. Preliminaries

2.1. **Results from the survey.** We use three results of [6], each proved there under (IID) and (EL).

- (R1) [6, Theorem 2.5] The range $\{X_n : n \geq 0\}$ is either P_0-almost surely finite or P_0-almost surely infinite.
- (R2) [6, Lemma 3.18] If the range is P_0-almost surely infinite, then for every $\ell \in \mathbb{R}^d \setminus \{0\}$ and all real $u < w$, the walk almost surely does not stay in the half-space $\{x : x \cdot \ell \leq w\}$ forever while visiting the slab $\{x : u \leq x \cdot \ell \leq w\}$ infinitely often.
- (R3) [6, Theorem 3.16] For every $\ell \in \mathbb{R}^d \setminus \{0\}$, $P_0(A_\ell \cup A_{-\ell}) \in \{0, 1\}$.

The survey states (R2) under (IID) and $\mathbb{E}[\omega(0, e, 1)] > 0$ for all $e \in E$, which (EL) implies. We use (R2) only with ℓ a coordinate vector and u, w integers.

Lemma 2.1. *Assume (IID) and (EL), suppose the range is P_0-almost surely infinite, and let $e \in E$. Then*

$$P_0\Big(\inf_n X_n \cdot e > -\infty,\ X_n \cdot e \not\to +\infty\Big) = 0.$$

Proof. On the event in question there are integers $a < b$ such that $X_n \cdot e \geq a$ for all n and $X_n \cdot e \leq b$ for infinitely many n. For fixed $a < b$ this is the event of (R2) with $\ell = -e$, $w = -a$ and $u = -b$, which is null. Take the union over the countably many pairs (a, b). □

2.2. **The counting lemma.** For a nearest-neighbour path $(x_n)_{n \geq 0}$ in $\mathbb{Z}^d$ with $x_0 \cdot e = 0$, a direction $e \in E$ and integers $m \geq 0$, $L \geq 1$, write

$$T_m = \inf\{n \geq 0 : x_n \cdot e = m\}, \qquad I_{m,L} = \{n : T_m \leq n < T_{m+L},\ x_n \cdot e = m\},$$

and, when $T_{m+L} < \infty$, $h_{m,L} = \max I_{m,L} - T_m$; otherwise $h_{m,L} = \infty$. Thus $h_{m,L}$ is the time from the first visit to level m to the last visit to level m before level $m + L$ is reached. For an integer $c \geq 0$ put

$$F_{M,L}(c) = \frac{1}{M+1} \sum_{m=0}^{M} \mathbf{1}\{h_{m,L} \leq c\}.$$

Since $h_{m,L}$ does not decrease in L, $F_{M,L}(c)$ does not increase in L. The next lemma is the deterministic part of Zerner's argument; it concerns a single path and its proof in [9] uses no property of the environment.

Lemma 2.2 ([9, Lemma 3])**.** *If $\limsup_n x_n \cdot e/n > 0$, then there is an integer $c \geq 0$ with*

$$\inf_{L \geq 1} \limsup_{M \to \infty} F_{M,L}(c) > 0.$$

3. The estimate at a new level

Fix $e \in E$. For an integer $c \geq 0$ let Π_c be the finite set of nearest-neighbour paths $p = (p_0, \ldots, p_s)$ with $p_0 = 0$, $s \leq c$ and $p_s \cdot e = 0$. For such a path let $k_p(x) = \#\{0 \leq t < s : p_t = x\}$ be the number of departures from x along p, and let $\omega^p(x, a, i) = \omega(x, a, i + k_p(x))$ be the environment with those entries removed. The endpoint p_s has not been left along p after its last visit, so its next entry is not removed. For $L > c$ define

$$q_L^p(\omega) = P_{p_s,\omega^p}\big(T_L < D\big), \qquad D = \inf\{n \geq 1 : X_n \cdot e \leq 0\},$$

where T_L is the hitting time of level L by the walk started at p_s. On $\{T_L < D\}$ the walk leaves level 0 at its first step and stays strictly above it until it reaches level L, so q_L^p depends only on the stacks at sites x with $x \cdot e \geq 0$, even if p itself visits negative levels. As L increases, q_L^p decreases to

$$q_\infty^p(\omega) = P_{p_s,\omega^p}\Big(X_n \cdot e > 0 \text{ for all } n \geq 1 \text{ and } \sup_n X_n \cdot e = \infty\Big),$$

the probability of reaching every level without returning to level 0. Put

$$a_L(c) = \sum_{p \in \Pi_c} \mathbb{E}\big[q_L^p\big], \qquad L > c.$$

Lemma 3.1. *Assume (IID) and (EL), suppose the range is P_0-almost surely infinite, and suppose $P_0(A_e) = 0$. Then for every $c \geq 0$, $a_L(c) \to 0$ as $L \to \infty$.*

Proof. Fix $p \in \Pi_c$ and let $w_p(\omega) = P_{0,\omega}\big((X_0, \ldots, X_s) = p\big)$, a finite product of entries of ω, which is positive $\mathbb{P}$-almost surely by (EL). After following p the walk sits at p_s with exactly the entries ω^p left, so by (1), $w_p\, q_\infty^p$ is the probability under $P_{0,\omega}$ that the walk follows p and afterwards stays strictly above level 0 and reaches every level. On this event $X_n \cdot e \geq -c$ for all n, so by Lemma 2.1 the event is contained in A_e up to a P_0-null set. Hence $\mathbb{E}[w_p q_\infty^p] \leq P_0(A_e) = 0$, and since $w_p > 0$ almost surely, $q_\infty^p = 0$ almost surely. By monotone convergence $\mathbb{E}[q_L^p] \to 0$, and Π_c is finite. □

Lemma 3.2. *Assume (IID). Let $m \geq 0$, $c \geq 0$ and $L > c$, let $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$, and let*

$$B_{m,L,c} = \{T_{m+L} < \infty,\ h_{m,L} \leq c\}.$$

Then $P_0(B_{m,L,c} \mid \mathcal{F}_{T_m}) \leq a_L(c)$ on $\{T_m < \infty\}$.

Proof. Fix a finite path $\eta = (\eta_0, \ldots, \eta_t)$ with $\eta_0 = 0$ that reaches level m for the first time at its last step, and let $x = \eta_t$. Before time t the walk has left only sites at levels below m, so the probability of η under $P_{0,\omega}$ is a product of entries of stacks at sites y with $y \cdot e < m$. By (IID), conditionally on $\{(X_0, \ldots, X_t) = \eta\}$ the stacks at sites y with $y \cdot e \geq m$ keep their original joint law.

On $B_{m,L,c} \cap \{(X_0, \ldots, X_t) = \eta\}$ let $t + s$ be the last visit to level m before T_{m+L}, so $s \leq c$, and let $p = (X_{t+r} - x)_{0 \leq r \leq s}$. Then $p \in \Pi_c$, and since $L > c$ the walk has not reached level $m + L$ by time $t + s$. The event is the disjoint union over $p \in \Pi_c$ of the events that the walk follows $x + p$ from time t and then reaches level $m + L$ before returning to level m. Given the environment, by (1) the probability of the second part, given the first, is $q_L^p(\theta_x \omega')$, where ω' is the environment with the entries used by η removed and $(\theta_x \omega')(z, a, i) = \omega'(x + z, a, i)$. As noted above, q_L^p reads only stacks at levels $\geq m$ after the shift, and these were not touched by η, so $q_L^p(\theta_x \omega') = q_L^p(\theta_x \omega)$. Writing $W_\eta^p(\omega) \leq 1$ for the conditional probability of following $x + p$, we get

$$P_{0,\omega}\big(B_{m,L,c} \,\big|\, (X_0, \ldots, X_t) = \eta\big) = \sum_{p \in \Pi_c} W_\eta^p(\omega)\, q_L^p(\theta_x \omega) \leq \sum_{p \in \Pi_c} q_L^p(\theta_x \omega).$$

The right side depends only on the stacks at levels $\geq m$. Averaging over the environment conditionally on η, these stacks have their original law, and by translation invariance of that law $\mathbb{E}[q_L^p \circ \theta_x] = \mathbb{E}[q_L^p]$. This gives the bound $a_L(c)$ for every η, which is the claim. □

4. Proof of Proposition 1.2

If the range is P_0-almost surely finite, then X_n is bounded and $X_n/n \to 0$, so by (R1) we may assume that it is almost surely infinite. Suppose $P_0(A_e) = 0$.

Fix $c \geq 0$, $L > c$ and a residue $r \in \{0, \dots, L-1\}$, and put $m_k = r + kL$, $\mathcal{G}_k = \mathcal{F}_{T_{m_k}}$ and $Z_k = \mathbf{1}_{B_{m_k,L,c}}$, with the convention that $\mathcal{F}_T = \mathcal{F}_\infty$ on $\{T = \infty\}$. The event $B_{m_k,L,c}$ is determined by the path up to $T_{m_k+L} = T_{m_{k+1}}$, so Z_k is $\mathcal{G}_{k+1}$-measurable, and $Z_k - E_0[Z_k \mid \mathcal{G}_k]$ is a bounded martingale difference sequence. By the strong law of large numbers for such sequences and Lemma 3.2,

$$\limsup_{K\to\infty} \frac{1}{K} \sum_{k<K} Z_k \leq a_L(c) \qquad P_0\text{-a.s.}$$

If some T_{m_k} is infinite then $Z_j = 0$ for all $j \geq k$ and the bound holds trivially. Averaging over the L residues and intersecting over the countably many pairs (c, L), almost surely, for all $c \geq 0$ and $L > c$,

$$\limsup_{M\to\infty} \frac{1}{M+1} \sum_{m=0}^{M} \mathbf{1}_{B_{m,L,c}} \leq a_L(c). \tag{2}$$

Now take a path in the almost sure event of (2) with $\limsup_n X_n \cdot e/n > 0$. Every level is reached, so $\mathbf{1}_{B_{m,L,c}} = \mathbf{1}\{h_{m,L} \leq c\}$ and the left side of (2) is $\limsup_M F_{M,L}(c)$. By Lemma 3.1, $\inf_{L>c} \limsup_M F_{M,L}(c) = 0$ for every c; since $F_{M,L}(c)$ does not increase in L, the infimum over all $L \geq 1$ is also zero. This contradicts Lemma 2.2. Hence $\limsup_n X_n \cdot e/n \leq 0$ almost surely. □

5. Proofs of Corollary 1.3 and Theorem 1.1

Proof of Corollary 1.3. If $P_0(A_e \cup A_{-e}) = 0$, then $P_0(A_e) = P_0(A_{-e}) = 0$. Proposition 1.2, applied to e and to $-e$, gives $\limsup_n X_n \cdot e/n \leq 0$ and $\liminf_n X_n \cdot e/n \geq 0$ almost surely. □

Proof of Theorem 1.1. If $P_0(N) = 0$ there is nothing to prove. Suppose $P_0(N) > 0$ and let $j \in \{1, \dots, d\}$. On $A_{e_j} \cup A_{-e_j}$ we have $|X_n| \to \infty$, so N is disjoint from it and $P_0(A_{e_j} \cup A_{-e_j}) < 1$. By (R3) this probability is 0, and Corollary 1.3 gives $X_n \cdot e_j/n \to 0$ almost surely. Hence $X_n/n \to 0$ P_0-almost surely, and in particular on N. □

Remark 5.1. The event N need not be the event that every site is visited infinitely often. For instance, in $d = 2$ give each site an independent uniformly chosen $a \in E$, and let the i-th entry of its stack put mass $1 - 3\varepsilon_i$ on a and $\varepsilon_i = 2^{-i-4}$ on each other direction. With positive probability the directions at 0 and e_1 point at each other and the walk moves back and forth between these two sites forever, since

$$\frac{1}{16} \prod_{i\geq 1} \left(1 - 3 \cdot 2^{-i-4}\right)^2 > 0.$$

This environment satisfies (IID) and (EL), and on that event the speed is zero, as Theorem 1.1 requires.

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

A Codex agent (OpenAI) proved Theorem 1.1; a second Codex agent checked the proof as a referee; a Claude agent (Anthropic) checked the steps and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code availability

Exact finite-horizon checks of the decomposition in Lemma 3.2, checks of the averages in Lemma 2.2 on random paths, and a simulation of the example in the closing remark are in the repository

`https://github.com/yspennstate/excited-walk-zero-speed-non-transience`.

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Appendix A. Notation

Symbol	Meaning	First appears
E	the $2d$ unit vectors of $\mathbb{Z}^d$	Section 1
$\omega(x,a,i)$	probability of step a on the i-th visit to x	Section 1
$P_{y,\omega}$, P_y, $\mathbb{P}$	quenched law, annealed law, law of the environment	(1)
A_ℓ	the event $X_n \cdot \ell \to +\infty$	Section 1
N	the event $\lvert X_n\rvert \not\to \infty$	Section 1
T_m, $I_{m,L}$, $h_{m,L}$	first hitting time of level m, visits to level m before level $m+L$, time from T_m to the last of them	Section 2
$F_{M,L}(c)$	fraction of levels $m \le M$ with $h_{m,L} \le c$	Section 2
Π_c, k_p, ω^p	short paths, their departure counts, the environment with their cookies removed	Section 3
q_L^p, q_∞^p, D	chance of reaching level L before returning to level 0, its limit, the return time	Section 3
$a_L(c)$	$\sum_{p\in\Pi_c} \mathbb{E}[q_L^p]$	Section 3
$B_{m,L,c}$, $\mathcal{F}_n$	the event $T_{m+L} < \infty$, $h_{m,L} \le c$; the natural filtration	Lemma 3.2
W_η^p, θ_x	probability of following $x+p$ after η; the shift by x	Lemma 3.2

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